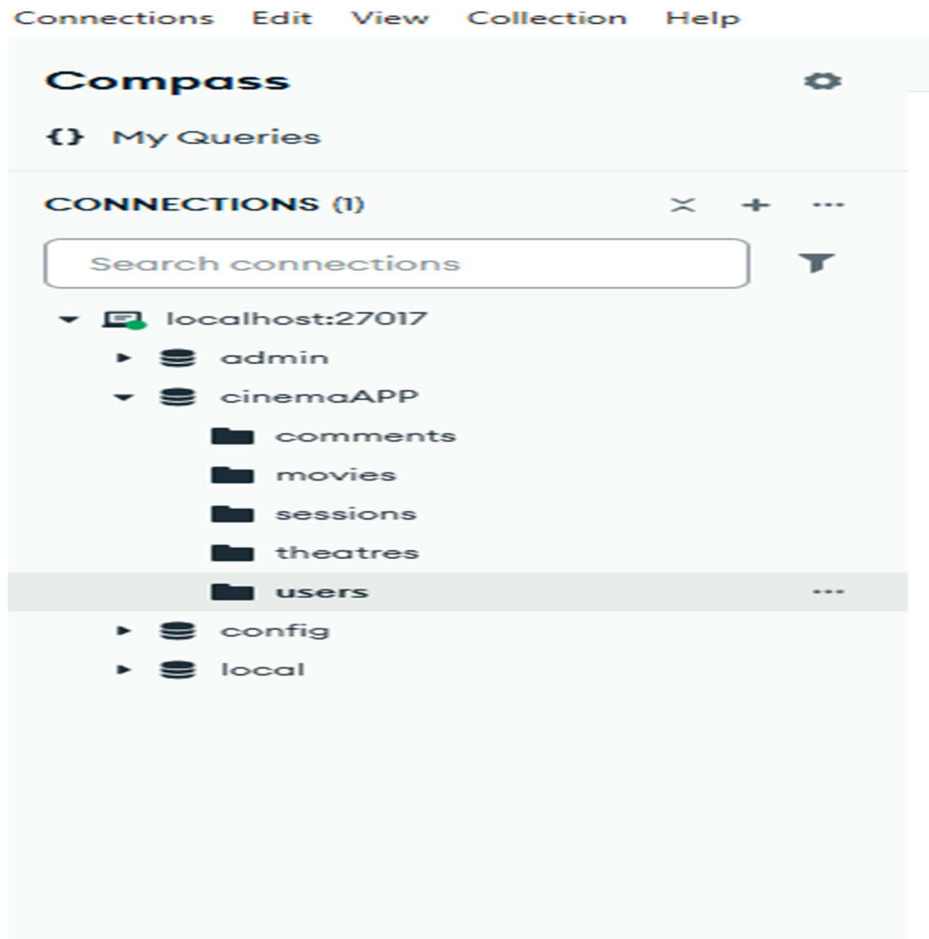


Tasca S5.01. Consultes amb MongoDB

Nivell 1

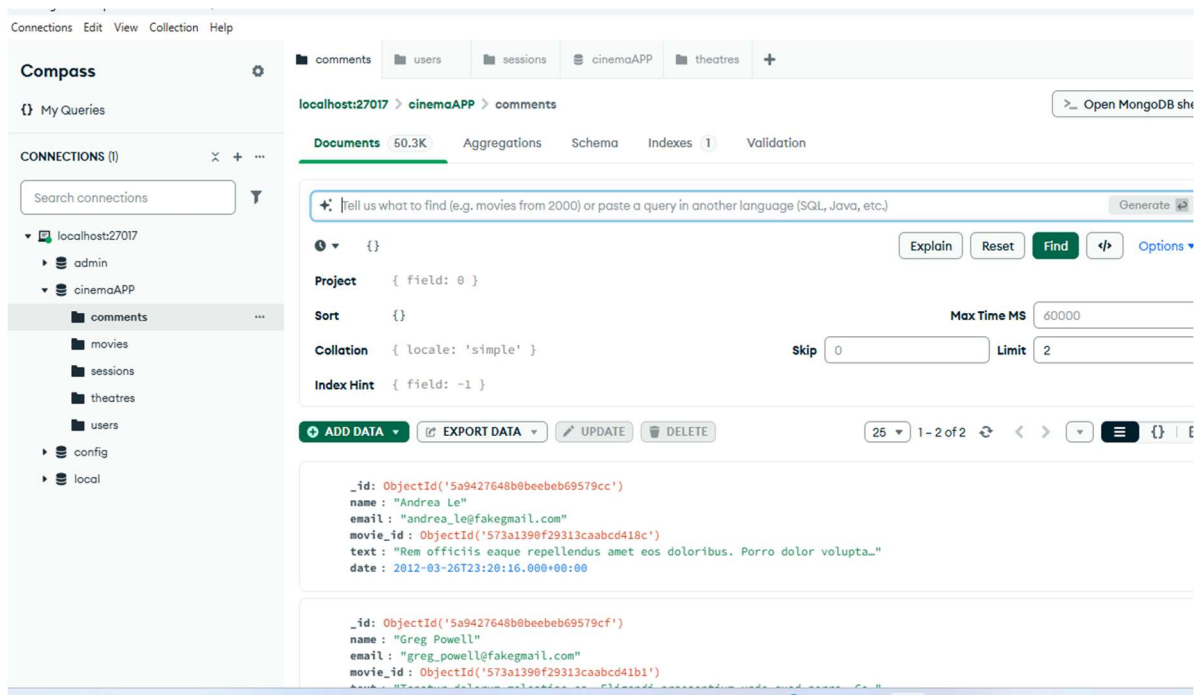
Crea una base de dades amb MongoDB utilitzant com a col·leccions els arxius adjunts.



Exercici 1

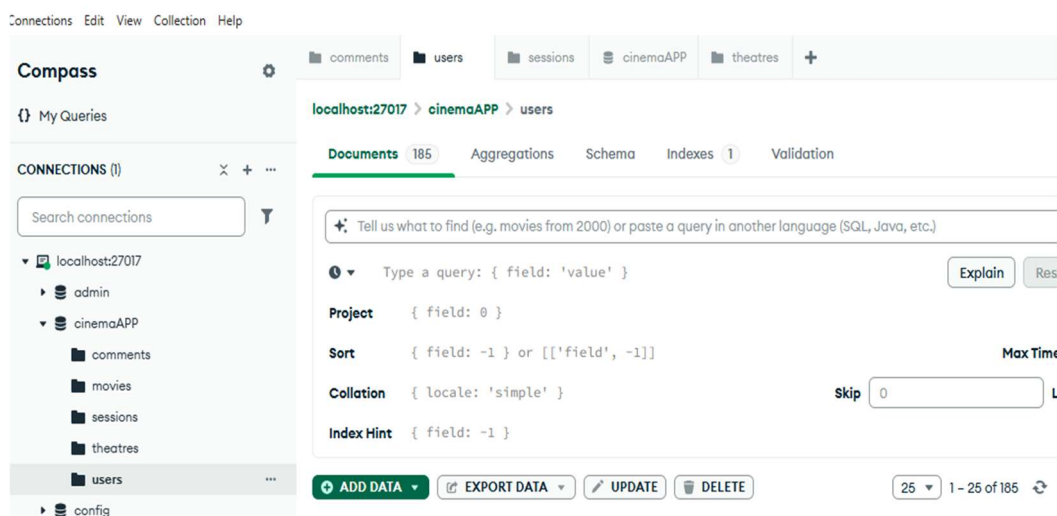
- **Mostra els 2 primers comentaris que hi ha en la base de dades.**

Anem, a col·lecció comments i a la pestanya documents i omplim els camps com i es veu en la captura (filtre{}, sort{} i límit 2).



○ Quants usuaris tenim registrats?

Anem, a col·lecció users i a la part superior de documents es veu els registrats (185).



○ Quants cinemes hi ha en l'estat de Califòrnia?

Anem, a col·lecció theatres i a la pestanya documents fem un filtre per califòrnia.

Connections Edit View Collection Help

Compass

My Queries

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- cinemaAPP
 - comments
 - movies
 - sessions
 - theatres
 - users
- config
- local

localhost:27017 > cinemaAPP > theatres

Documents 1.6K Aggregations Schema Indexes 1 Validation

Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.)

Find

Project { field: 0 }

Sort { field: -1 } or [['field', -1]]

Collation { locale: 'simple' }

Index Hint { field: -1 }

Max Time MS 60000

Skip 0 Limit 0

ADD DATA EXPORT DATA UPDATE DELETE

25 1 - 25 of 169

```

{
  "_id": {},
  "theaterId": 1011,
  "location": {}
}

```

- Quin va ser el primer usuari/ària en registrar-se?

Connections Edit View Collection Help

Compass

My Queries

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- cinemaAPP
 - comments
 - movies
 - sessions
 - theatres
 - users
- config
- local

localhost:27017 > cinemaAPP > users

Documents 185 Aggregations Schema Indexes 1 Validation

Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.)

Find

Project { field: 0 }

Sort { "_id": 1 }

Collation { locale: 'simple' }

Index Hint { field: -1 }

Max Time MS 60000

Skip 0 Limit 1

ADD DATA EXPORT DATA UPDATE DELETE

25 1 - 1 of 1

#	_id ObjectId	name String	email String	password String
1	ObjectId('59b99db4cfa9a3...	"Ned Stark"	"sean_bean@gameofthron.e...	"\$2b\$12\$UREFwsRUoyF0CRqG...

- **Quantes pel·lícules de comèdia hi ha en la nostra base de dades?**

Anem a col·lecció movies i a la pestanya documents fem un filtre per genres : Comedy.

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists 'localhost:27017' with databases 'admin', 'cinemaAPP', and 'local'. The 'cinemaAPP' database is expanded, showing collections 'comments', 'movies', 'sessions', 'theatres', and 'users'. The 'movies' collection is selected. The main panel shows the 'Documents' tab for 'localhost:27017 > cinemaAPP > movies'. A query filter is applied: `{ "genres": "Comedy" }`. The document viewer shows a single document for 'Gertie the Dinosaur' with the following fields: `_id`, `plot`, `genres` (array with 'Animation', 'Short', 'Comedy'), `runtime`, `cast`, `num_mflix_comments`, `poster`, `title`, `fullplot`, `languages`, `released`, `directors`, `writers`, and `awards`.

Exercici 2

Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estigui en francès.

Faig un consulta en 2 condicions: l'any de producció (1932) i el gènere (drama) o país (França).

The screenshot shows the MongoDB Compass interface with the same setup as before. The query filter is updated to: `{ "year": 1932, "$or": [ { "genres": "Drama" }, { "countries": "France" } ] }`. The document viewer shows a single document for 'The Blue Light' with the following fields: `plot`, `genres` (array with 'Drama', 'Short', 'Comedy'), `runtime`, `cast`, `poster`, `title`, `fullplot`, `languages`, `released`, `directors`, `writers`, `awards`, `lastupdated`, and `year`.

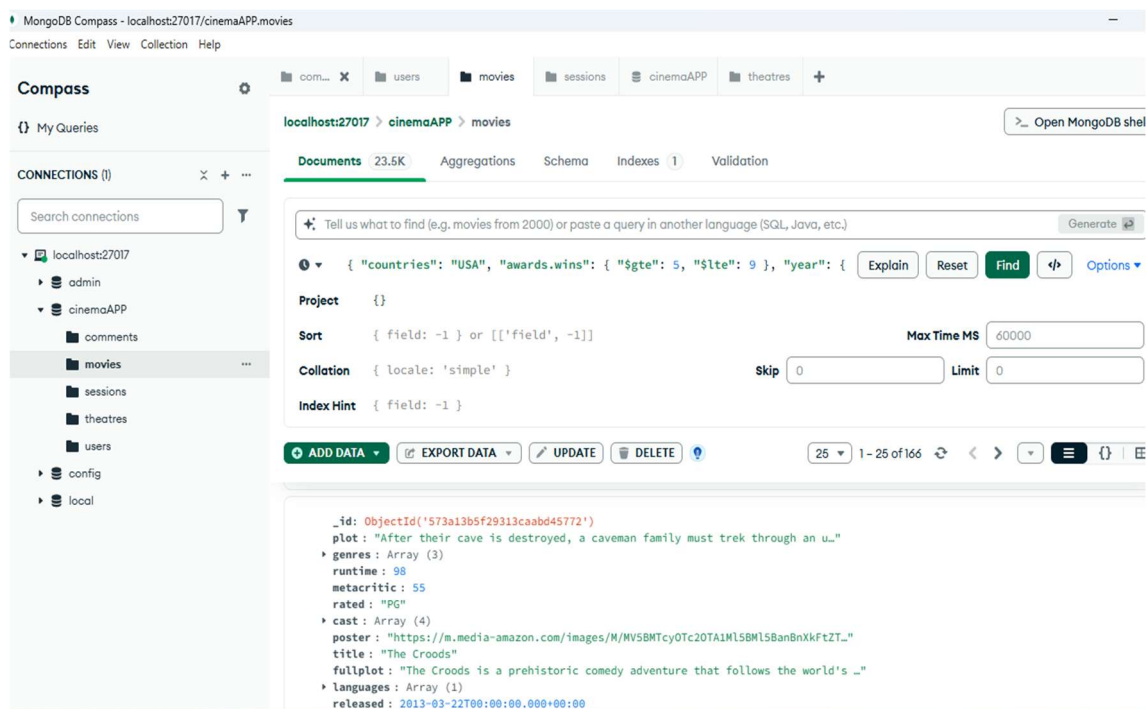
Exercici 3

Mostra'm tots els documents de pel·lícules estatunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

Aquí ens demanen 3 condicions: País (USA), Premis (dintre rang 5 i 9) i any (entre 2012 i 2014).

No es veu massa complert i he fet:

```
{ "countries": "USA", "awards.wins": { "$gte": 5, "$lte": 9 }, "year": { "$gte": 2012, "$lte": 2014 } }
```



Nivell 2

Exercici 1

Compte quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

Utilitzo pipeline d'agregació per filtrar per aquest correu i després comptar els comentaris.

Compass

{ My Queries

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- cinemaAPP
 - comments
 - movies
 - sessions
 - theatres
 - users
- config
- local

localhost:27017 > cinemaAPP > comments

Documents 50.3K Aggregations Schema Indexes 1 Validation

Tell us what aggregation to build (e.g. count movies made each year) or paste one in another language (SQL, Java, etc.)

Generate

\$match \$count

Explain Export Run Options

Untitled SAVE + CREATE NEW EXPORT TO LANGUAGE PREVIEW STAGES TEXT WIZARD

Stage 1 \$match

```

1
2 {
3   "email": {
4     "$regex": "@GAMEOFTHRON\\.ES$",
5     "$options": "i"
6   }
7 }
8

```

Output preview after \$match stage (Sample of 10 documents)

_id: ObjectId('5a9427648b0beeb69579d0')	_id: ObjectId('5a9427648b0beeb69579d0')
name: "Talisa Maegyr"	name: "Petyr"
email: "oona_chaplin@gameofthron.es"	email: "aida"
movie_id: ObjectId('573a1390f29313caabcd41b...')	movie_id: ObjectId('573a1390f29313caabcd41b...')
text: "Rem itaque ad sit rem voluptatibus. Ad fugiat maxime illum optio iure ..."	text: "Quo d nemo n"
date: 1998-08-22T11:45:03.000+00:00	date: 2001-08-22T11:45:03.000+00:00

Stage 2 \$count

En 2 parts:

Connections Edit View Collection Help

Compass

{ My Queries

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- cinemaAPP
 - comments
 - movies
 - sessions
 - theatres
 - users
- config
- local

localhost:27017 > cinemaAPP > comments

Documents 50.3K Aggregations Schema Indexes 1 Validation

Tell us what aggregation to build (e.g. count movies made each year) or paste one in another language (SQL, Java, etc.)

Generate

\$match \$count

Explain Export Run Option

Untitled SAVE + CREATE NEW EXPORT TO LANGUAGE PREVIEW STAGES TEXT WIZARD

Stage 2 \$count

```

1 "totalComentaris"
2

```

Output preview after \$count stage (Sample of 1 document)

```
totalComentaris : 22841
```

Resultat:

MongoDB Compass - localhost:27017/cinemaAPP.comments

Connections Edit View Collection Help

Compass

My Queries

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- cinemaAPP
 - comments
 - movies
 - sessions
 - theatres
 - users
- config
- local

localhost:27017 > cinemaAPP > comments

Documents 50.3K Aggregations Schema Indexes 1 Validation

Match Count Edit

Collation { locale: 'simple' } Max Time MS 60000

ALL RESULTS Showing 1 - 1 count results

totalComentarios : 22841

Comprovació:

MongoDB Compass - localhost:27017/cinemaAPP.comments

Connections Edit View Collection Help

Compass

My Queries

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- cinemaAPP
 - comments
 - movies
 - sessions
 - theatres
 - users
- config
- local

localhost:27017 > cinemaAPP > comments

Documents 50.3K Aggregations Schema Indexes 1 Validation

Query: { "\$regex": "@GAMEOFTHRON\\.ES\$", "\$options": "i" } Generate query Explain Reset Find Options

Project { field: 0 }

Sort {} Max Time MS 60000

Collation { locale: 'simple' } Skip 0 Limit 0

Index Hint { field: -1 }

ADD DATA EXPORT DATA UPDATE DELETE

25 1 - 25 of 22841

name: "Jorah Mormont"
email: "iain_glen@gameofthron.es"
movie_id: ObjectId('573a1390f29313caabed44d3')
text: "Minus sequi incidunt cum magnam. Quam voluptatum vitae ab voluptatum c..."
date: 1994-02-18T18:52:31.000+00:00

_id: ObjectId('5a9427648b08beebe69579f7')
name: "Yara Greyjoy"
email: "emma_wheeler@gameofthron.es"
movie_id: ObjectId('573a1390f29313caabed4556')
text: "Dignissimos sunt aspernatur rerum magni debitis neque. Temporibus nisi..."
date: 2016-10-05T16:26:16.000+00:00

Exercici 2

Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

Aquí primer filtro i després agrupo per "estat"(location.address.zipcode)

The screenshot shows the MongoDB Compass interface. The left sidebar displays the database structure with 'cinemaAPP' selected, and 'theatres' is the current collection. The main panel shows an aggregation pipeline with two stages: '\$match' and '\$group'. The '\$match' stage has a query: `{ "location.address.state": "DC" }`. The '\$group' stage has a query: `{ "_id": "$location.address.zipcode", "numCinemes": { "$sum": 1 } }`. The output preview for the '\$match' stage shows a sample of 3 documents. The output preview for the '\$group' stage shows a sample of 3 documents, with the first one being: `{ "_id": "20002", "numCinemes": 1 }`.

Resultats:

The screenshot shows the MongoDB Compass interface with the aggregation pipeline executed. The main panel displays the results of the aggregation query. The results are shown in a table with 3 rows, each representing a zip code and the number of cinemas (numCinemes) located there. The results are: `{ "_id": "20002", "numCinemes": 1 }`, `{ "_id": "20010", "numCinemes": 1 }`, and `{ "_id": "20016", "numCinemes": 1 }`.

Nivell 3

Exercici 1

Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

Faig una **pipeline d'agregació** a MongoDB, per **seleccionar i mostrar només certs camps** dels documents: pel·lícules dirigides per John Landis i puntuació (IMDb) entre 7,5 i 8.

The screenshot shows the MongoDB Compass interface. The left sidebar displays the database structure with 'cinemaAPP' selected, and the 'movies' collection highlighted. The main panel is in the 'Aggregations' tab, showing a single stage named '\$match'. The aggregation pipeline is defined as:

```
1 {
2   "directors": "John Landis",
3   "imdb.rating": {
4     "$gte": 7.5,
5     "$lte": 8
6   }
7 }
```

The output preview on the right shows a sample of 4 documents after the '\$match' stage. The first document is:

```
{
  "_id": ObjectId('573a1397f29313caabce6d94'),
  "fullplot": "Faber College has one frat house so disreputable it will take anyone. _",
  "imdb": {
    "year": 1978,
    "plot": "At a 1962 college, Dean Vernon Wormer is determined to expel the entire..."
  },
  "genres": "Arra",
  "runtime": 133,
  "rated": "R",
  "cast": Array,
  "num_flix_con": ...
}
```

En 2 parts:

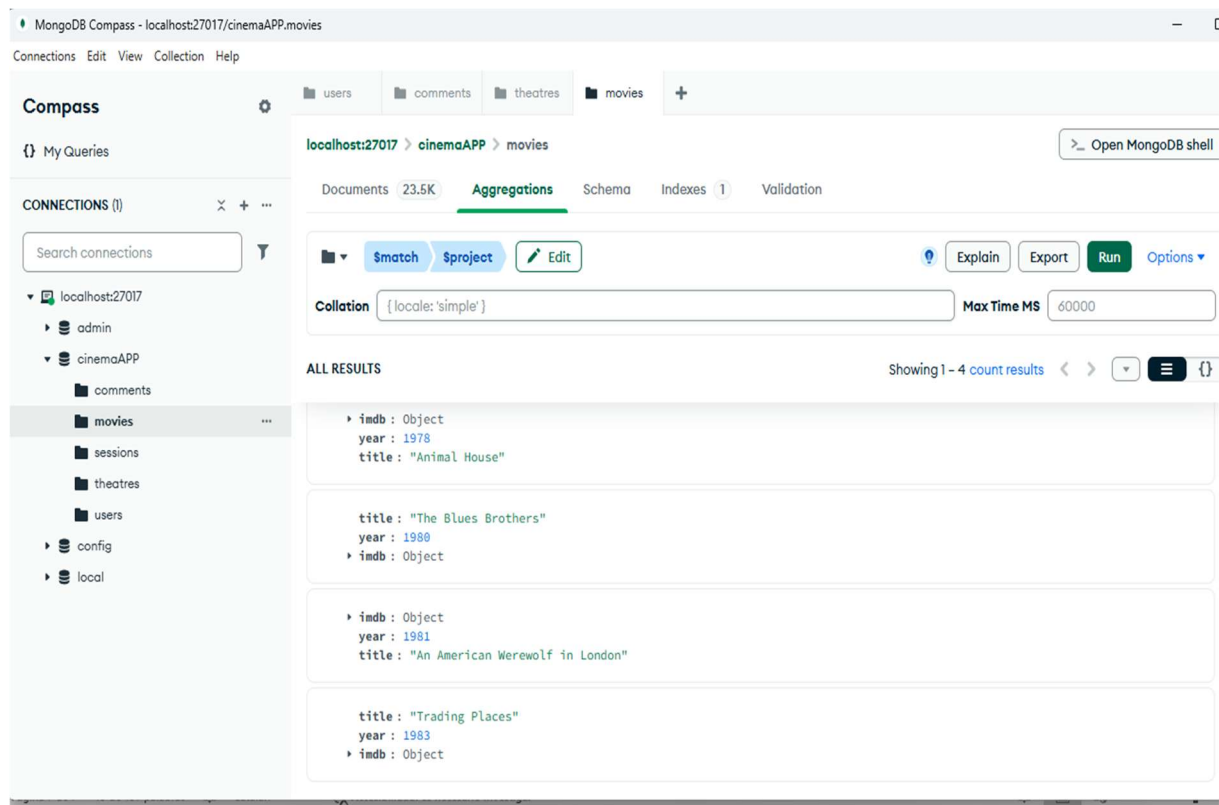
The screenshot shows the MongoDB Compass interface with the aggregation pipeline updated to include a '\$project' stage. The pipeline is now:

```
1 {
2   "title": 1,
3   "year": 1,
4   "imdb.rating": 1,
5   "_id": 0
6 }
```

The output preview on the right shows a sample of 4 documents after the '\$project' stage. The first document is:

```
{
  "imdb": {
    "year": 1978,
    "title": "Animal House"
  },
  "title": "The",
  "year": 1980,
  "imdb": {
    "year": 1980,
    "title": "The"
  }
}
```

Resultat:



Exercici 2

Mostra en un mapa la ubicació de tots els teatres de la base de dades.

Ens assegurem que cada document té el camp:

